# MBDyn Installation Manual
# Version develop

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# Chapter 1

# Introduction

This document describes how to download, build, install and execute MBDyn — MultiBody Analysis program, a suite of tools for multibody/multidisciplinary analysis of complex systems.

For any question or problem, to fix typos, bugs, for comments and suggestions, please contact the Development Team without hesitation:

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This document is also available online at
http://www.aero.polimi.it/~masarati/MBDyn-input/install/

# Chapter 2

# Getting the package

## 2.1 MBDyn

The package can be downloaded in source form from

`http://www.aero.polimi.it/~mbdyn/`

Binary releases and snapshots are also available for Windows 2000/XP at `http://www.aero.polimi.it/~masarati/Download/mbdyn/`, compiled with Cygwin (see `http://www.cygwin.com/`) and thus require `cygltdl-3.dll`; they are provided to Windows users to save them the burden of installing Cygwin (very easy and straightforward, though) and to compile a package under an unfamiliar environment. However, no support is provided for those builds, unless problems are easily identifiable as related to the sources and not to the OS, and they impact the UN*X version as well.

MBDyn may use a wide range of packages, if available on the host system and correctly detected by `configure`.

## 2.2 Mathematical Utilities

MBDyn may exploit the availability of some mathematical utilities; neither of them is required for a basic compilation, but they may be useful or required for specific features.

### 2.2.1 ATLAS

The Automatically Tunable Linear Algebra Subroutines are a replacement of the standard BLAS. They are exploited by the linear solver Umfpack (see Section 2.3.3) and can be used by Lapack (see Section 2.3.4) and other packages that require BLAS (see Section 2.2.2).

Note that ATLAS provides its own implementation of Lapack but only for a limited set of functions. This set of functions is packed into an archive called

`liblapack.a`, so in typical installations it may shadow complete implementations of the Lapack package. A useful "trick" to have a complete Lapack installation that exploits ATLAS performances where available is described in `http://math-atlas.sourceforge.net/errata.html#completelp`. The essential instructions are reported below, assuming that `$ATLAS` and `$LAPACK` respectively are the paths to your ATLAS and Lapack static library archive files:

```
mkdir tmp
cd tmp
ar x $ATLAS/liblapack.a
cp $LAPACK/liblapack.a $ATLAS/liblapack.a
ar r $ATLAS/liblapack.a *.o
cd ..
rm -rf tmp
```

### 2.2.2 BLAS

The Basic Linear Algebra Subroutines can be used by Umfpack (see Section 2.3.3) and other packages. Specially tuned binaries for each architecture (processor type, cache size and other special hardware features) are advisable; otherwise, instead of compiling them of your own, ATLAS (see Section 2.2.1) are considered a better replacement.

## 2.3 Linear Solvers

MBDyn may use a variety of linear solvers

### 2.3.1 Naive

The Naive solver is also native. It is especially meant for medium size problems (between 100 and 1000 unknowns) and is essentially a sparse solver optimized for speed rather than memory usage.

### 2.3.2 Y12

The Y12 solver is also built-in. the MBDyn Project distributes the Y12 library **AS IS** and **WITHOUT ANY WARRANTY** as part of the source code, with no copyright statement and no license. Of course credits go to the original Authors: Zahari Zlatev, Jerzy Wasniewski, and Kjeld Schaumburg, Comp. Sci., Math. Inst., University of Aarhus, Ny Munkegade, DK 8000 Aarhus. This solver is recommended for moderate to large problems, if Umfpack (see Section 2.3.3) is not available.

### 2.3.3 Umfpack

The Umfpack linear sparse solver library must be downloaded separately, from `http://www.cise.ufl.edu/research/sparse/umfpack/`. Credit goes to Timothy A. Davis, University of Florida. Umfpack is used by permission; please read its Copyright, License and Availability note. It is used as the standard sparse solver by MATLAB (see `http://www.mathworks.com/`). This solver is recommended for very large problems, and as a general purpose solver.

### 2.3.4 Lapack

### 2.3.5 SuperLU (experimental)

### 2.3.6 TAUCS (experimental)

### 2.3.7 Harwell (historical)

## 2.4 Utilities

## 2.5 Communication

### 2.5.1 MPI

### 2.5.2 Metis

Metis is a package that performs automatic domain decomposition. It is used by the Schur solution manager to partition the model into submodels of nearly equal computational cost and with minimal interface size. Its compilation is straightforward. It can be downloaded from `http://www-users.cs.umn.edu/~karypis/metis/metis/`. Its compilation for MBDyn used to require a special patch to cleanup its namespace; now the patch is no longer required as a better namespace separation has been operated. No installation procedure is provided as of version 4.0; at the end of compilation, the library `libmetis.a` is available in the build tree, and it should be copied to a directory where the loader can locate it; the headers `Lib/*.h` should be copied where the C preprocessor can locate them.

## 2.6 Real-Time

### 2.6.1 RTAI

## 2.7 Build Environments

### 2.7.1 GNU/Linux

The preferred build environment is GNU/Linux, using gcc, g++ and either g77 or gfortran.

### 2.7.2 Windows: CygWin

MBDyn has been successfully compiled under CygWin. No special requirement is known. Special attention is required to build with some specific feature, significantly run-time loadable modules.

### 2.7.3 Windows: MSYS/MinGW

MBDyn has been successfully compiled under MSYS/MinGW. Special attention needs to be played to what version is used.

Download packages from `http://www.mingw.org/`.

Required packages (in order of installation):

- `MinGW-3.1.0-1.exe` (install wizard)
- `MSYS-1.0.10.exe` (install wizard; if not automagically done, edit file `/etc/fstab` as indicated in MSYS' documentation)
- `msysDTK-1.0.1.exe` (install wizard)
- `gcc-core-3.4.2-20040916-1.tar.gz` (archive; untar after changing directory into `/mingw`)
- `gcc-g++-3.4.2-20040916-1.tar.gz` (archive; untar after changing directory into `/mingw`)
- `gcc-g77-3.4.2-20040916-1.tar.gz` (archive; untar after changing directory into `/mingw`)
- `msys-libtool-1.5.tar.bz2` (archive; untar after changing directory into `/`)

### 2.7.4 PowerPC: Mac OS X

(Reported by Torsten Sadowski, blindly reported here)

You have to tell `configure` where to find `fink` includes and libraries

```
export LDFLAGS=-L/sw/lib
export CPPFLAGS=-I/sw/include
```

# Chapter 3

# Building

The configuration of the MBDyn package is based on GNU's autotools (see `http://www.gnu.org/` for details).

## 3.1 Configuring

### 3.1.1 Option List

Specific options (from `configure --help`):

```
--enable-debug          enable debugging (no)
--with-debug-mode[={none|mem}]  with debug mode {none|mem} (none)
--enable-socket-drives  enable socket drives (auto)
--enable-runtime-loading    enable runtime loading (auto)
--with-static-modules   build (known) modules as static (auto)
--enable-crypt          enable crypt (deprecated) (no)
--enable-schur          enable Schur parallel solver
                        (needs MPI and either Metis or Chaco) (auto)
--enable-multithread    enable multithread solution (no)
--with-tcl              with tcl interpreters (auto)
--with-libf2c[={f2c|g2c}] with f2c library (auto)
--with-fs[={unix|dos}]  filesystem type (unix)
--with-mpi              with MPI support (=pmpi for profiling) (auto)
--enable-debug-mpi      enable MPI debugging (no)
--with-metis            with Metis model partitioning support (auto)
--with-threads          with threads (auto)
--with-rtai             with RTAI support (no)

     math libraries:
--with-blas             with (C)BLAS math library (auto)
--with-goto=lib(s)      with Goto BLAS implementation
--with-ginac            with GiNaC support
                        (ginac-config must be in $PATH) (auto)
```

```
     linear algebra solvers (naive is enabled by default):
--with-y12              with Y12 sparse math library (yes)
--with-umfpack          with Umfpack math library (auto)
--with-lapack           with LAPACK math library (auto)
--with-harwell          with HSL (Harwell) sparse math library - historical (auto)
--with-superlu          with SuperLU math library - eXperimental (auto)

     misc security libraries:
--with-pam              with PAM support (auto)
--with-sasl2            with Cyrus SASL2 support (auto)

     supported features:
--with-struct           with structural elements (yes)
--with-elec             with electric stuff (yes)
--with-aero             with aerodynamic stuff (yes)
--with-aero-output={std,gauss,node}  aerodynamic output mode (auto)
--with-hydr             with hydraulic stuff (yes)

--with-module=<list>    build listed modules (see modules/)
```

### 3.1.2 Multithread Assembly/Solution

The switch `--enable-multithread` enables multithreaded assembly; it defaults to `auto`, i.e. if the system meets the following requirements, it is automatically enabled. Essentially, a working `-lpthread` library and a POSIX compliant `pthread.h` header must be available. Currently, only the SuperLU sparse threaded solver is available; it is experimental. A threaded implementation of the built-in `naive` sparse solver is under development.

### 3.1.3 Schur Parallel Solver

The Schur parallel solver is enabled by using the switch `--enable-schur`. It requires a working MPI library with the `ch` driver and the C++ interface, and a partitioning library. The MPI library is selected by the switch `--with-mpi`, which allows to specify the value `pmpi` if the profiling version of the library is to be used. Note that recent MPI releases by default only build the profiling version of the C++ library, so the `pmpi` value is mandatory. Currently, the only partitioning library that is supported by MBDyn is METIS; a patch to support Chaco is being incorporated, and may be available in future releases. The Schur solver is not compatible with the multithreaded assembly/solution, so if no switch is specified and the system meets the requirements for both, the multithreaded assembly wins over the Schur solver. As a consequence, the schur solver should be explicitly enabled.

### 3.1.4 Real-Time Simulator

The real-time simulator is enabled by using the switch `--with-rtai`, which essentially detects the availability of the mandatory headers of the GNU/Linux

Real-Time Application Interface. These headers changed between 2.4.13 and 3.X; both versions are automatically detected.

# Chapter 4

# Installing

Run `make install`; this essentially installs the binary, the utilities, the man page and the dynamic modules, if any.

# Chapter 5

# Executing

Prepare an input file and run `mbdyn -f <input> -o <output>`.

## 5.1 Regular Execution

## 5.2 Parallel Execution

## 5.3 Real-Time Execution

## 5.4 External Execution

To run MBDyn as a Simulink module, use the `SimulinkInterface` that is available under `contrib/`.

# Chapter 6

# Troubleshooting

# Chapter 7

# HOWTOs

This chapter contains mini-howtos about typical significant configurations of MBDyn as performed by the developers. Feel free to contribute your own if you had to do any unusual configuring to meet special needs.

## 7.1 Run-Time Modules

To enable run-time loading of modules, a powerful means to extend MBDyn functionality, one needs to:

- install, and let MBDyn detect, the ltdl library, an ancillary library of libtool that handles platform-independent run-time loading of software modules;
- enable run-time loading of modules; this requires to configure MBDyn with `--enable-runtime-loading`;
- when using gcc, it is recommended to add `-rdynamic` to the `LDFLAGS` environment variable; this is required to make functions and objects provided by MBDyn available in the modules;
- when developing a custom module, it must be placed in a subdirectory of the directory `modules/`, named `module-name/`. The module should be contained in one file, with the same name of the directory plus the language-specific extension; this should be
  - C: `module-name.c`;
  - C++: `module-name.cc`;
  - Fortran 77: `module-name.f`.
- the building of each module must be explicitly enabled; this requires to configure with `--with-module=list`, providing a list of the module names (without the leading `module-`).

For example, to enable run-time loading and to build the `wheel2` module, use

```
$ ./configure --enable-runtime-loading --with-module=wheel2
```

Occasionally, one may want to statically build some well-known modules into MBDyn, specifically those that implement new elements. This requires to configure with `--with-static-modules`. After that, the user-defined elements are available as joints. Currently, only the `wheel2` module can be statically built into MBDyn; this feature will require some reworking.

Run-time loading of dynamic modules is known to work on any flavor of GNU/Linux on X86/X86_64 and on Windows using Cygwin, when compiled with gcc. It may work on other architectures and with other compilers, but it has not been tested by, nor reported to, the Developers.

## 7.2 Schur Solver

This section describes how the Schur parallel solver available within the MBDyn package has been successfully compiled and executed on a dual Athlon SMP machine. By no means it is intended to suggest how the related packages should be built for other purposes, nor it may represent a replacement for the original build and install procedures. Please refer to the documentation available with each package for more details or for troubleshooting.
Software prerequisites:

- gcc/g++/g77 >= 3.0 (tested with gcc/g++/g77 3.2.1, 3.3.1 and 3.4.0)

Software requirements:

- MBDyn 1.2.1
- mpich 1.2.5.2
- Metis 4.0

**MPI**

- download `mpich.tar.gz` from `http://www-unix.mcs.anl.gov/mpi/`
- untar the package in a temporary directory
- configure the package with

  ```
  $ ./configure -rsh=ssh --disable-f77 --prefix=/usr/local/mpich
  ```

- `make` the package
- `make install`
- edit `/usr/local/mpich/share/machines.<arch>` to enumerate the max number instances of the MBDyn process you want to allow on each machine; in my case `<arch>` is LINUX (more details in `/usr/local/mpich/doc/mpichman-chp4.pdf`).

**Metis**

- download XXX from `http://www-users.cs.umn.edu/~karypis/metis/metis/`
- untar it in a temporary directory
- apply the patch `metis-namespace-cleanup.patch` from `http://www.aero.polimi.it/~masarati/Download/mbdyn/`
- `make` the package
- ...
- copy the library `libmetis.a` in `/usr/local/lib` and the header files `defs.h`, `macros.h`, `metis.h`, `proto.h`, `rename.h` and `struct.h` in `/usr/local/include/metis`.

**MBDyn**

- download `mbdyn-1.2.1.tar.gz` from `http://www.aero.polimi.it/~masarati/Download/mbdyn/`
- untar it in a temporary directory
- make sure the compiler can find headers from MPI and Metis by defining

```
CPPFLAGS="-I/usr/local/mpich/include \
    -I/usr/local/mpich/include/mpi2c++ \
    -I/usr/local/include/metis"
```

- make sure the linker can find the libraries from MPI and Metis by defining

```
LDFLAGS="-L/usr/local/mpich/lib \
    -L/usr/local/lib"
```

- configure MBDyn by running

```
$ ./configure --with-mpi=pmpi --with-metis --enable-schur \
    --prefix=/usr/local
```

- `make` the package
- `make install` the package

**Run It!**
To test the system, one needs a test input file; the example `cantilever2` from `http://www.aero.polimi.it/~mbdyn/documentation/examples/` should do the trick. The input file does not need any specific change, unless special features are required. To run it on an SMP machine, simply execute

```
$ /usr/local/mpich/bin/mpirun -np 2 mbdyn -f cantilever2 -o /tmp/ -
ss
```

This generates a set of files `/tmp/cantilever2.0.*` and `/tmp/cantilever2.1.*` with the outputs from processes 0 and 1.

## 7.3 Real-Time Simulator

This section describes how the Real-Time simulator available within the MBDyn package has been successfully compiled and executed. By no means it is intended to suggest how the related packages should be built for other purposes, nor it may represent a replacement for the original build and install procedures. Please refer to the documentation available with each package for more details or for troubleshooting.
Software prerequisites:

- gcc/g++/g77 >= 3.0 (tested with gcc 3.3.1 and 3.4.0)

Software requirements:

- MBDyn 1.2.1
- RTAI 3.0

**RTAI**

- download `rtai-3.0.tar.gz` from `http://www.rtai.org/`
- untar the package in a temporary directory
- configure the package with

  ```
  $ ./configure
  ```

- `make` the package
- `make install`

**MBDyn**

- download `mbdyn-1.2.1.tar.gz` from `http://www.aero.polimi.it/~masarati/Download/mbdyn/`
- untar it in a temporary directory
- make sure the compiler can find headers from RTAI by defining

```
CPPFLAGS="-I/home/realtime"
```

- configure MBDyn by running

```
$ ./configure --with-rtai
```

- `make` the package
- `make install` the package

**Run It!**
To test the system, one needs a test input file; the example `RT-MBDyn/pendulum` from `http://www.aero.polimi.it/~mbdyn/documentation/examples/` should do the trick. Simply execute

```
$ mbdyn -f pendulum -ss
```

...

## 7.4 Simulink Interface

This section describes how the Simulink Interface available within the MBDyn package has been successfully compiled and executed. By no means it is intended to suggest how the related packages should be built for other purposes, nor it may represent a replacement for the original build and install procedures. Please refer to the documentation available with each package for more details or for troubleshooting. An analogous interface with Scicos [1] is described in Section 7.5.
Software prerequisites:

- gcc/g++/g77 >= 3.0 (tested with gcc 3.3.1 and 3.4.0)
- Matlab/Simulink (tested with XXX)

Software requirements:

- MBDyn 1.2.1

**MBDyn**

- download `mbdyn-1.2.1.tar.gz` from `http://www.aero.polimi.it/~masarati/Download/mbdyn/`
- untar it in a temporary directory
- configure MBDyn by running

```
$ ./configure
```

- `make` the package
- `make install` the package
- in directory `contrib/SimulinkInterface`, edit the file `Makefile` such that the appropriate `mex` compiler is used; in my system it is `/opt/matlab/bin/mex`
- `make` the package

**Run It!**
To test the system, one needs a test input file; the example `pendulum` in the subdirectory `examples` should do the trick.

- start Matlab
- add the subdirectory `contrib/SimulinkInterface` to Matlab's path, by executing

```
> path('<path/to>/contrib/SimulinkInterface', path);
```

- execute simulink by clicking on the related icon, or by running

```
> simulink
```

- open the model by clicking `File->Open` on the menubar
- run the model by clicking `Simulation->Start` on the menubar. some times, the first run fails; we're still trying to track that down. In case, just try again...
- for more details on creating your own model, or on editing the interface parameters, refer to the `README` in `contrib/SimulinkInterface`

## 7.5 Scicos Interface

This section describes how the Scicos [1] interface available within the MBDyn package has been successfully compiled and executed. By no means it is intended to suggest how the related packages should be built for other purposes, nor it may represent a replacement for the original build and install procedures. Please refer to the documentation available with each package for more details or for troubleshooting.

An analogous interface with Simulink [2] is described in Section 7.4.

# Chapter 8

# Developers

## 8.1 Prepare for Building

Developers need to prepare the build environment by running

```
$ ./bootstrap.sh
```

This need appropriate versions of the autotools:

- `automake`: there seems to be no special version requirement;
- `autoconf`: there seems to be no special version requirement;
- `libtool`: needs to be at least version 1.5.

# Bibliography

[1] SCICOS. http://www.scilab.org/.

[2] Simulink. http://www.mathworks.com/.